API Business Analysis & Service Contracts — WariMitra

Document Identifier: WM-API-012

Classification: Technical BA Service Catalog & Interface Specifications

Architecture Pattern: Serverless Microservices (Firebase Cloud Functions v2)

Runtime Environment: Node.js 20 / TypeScript 5.3 (`asia-south1` Mumbai)

Target Audience: Technical Business Analysts, Integration Engineers, Solution Architects, API Evaluators

Source Evidence: `functions/src/index.ts`, `ARCHITECTURE.md`, `mobile/lib/services/`

1. Architectural Interface Paradigm

WariMitra implements a **hybrid serverless integration model**:

1. **HTTPS Callable Cloud Functions:** Structured RPC endpoints handling mission-critical business validations (idempotency, rate limiting, geospatial calculations) requiring elevated administrative privileges.

2. **Firestore Real-Time Event Triggers:** Asynchronous event-driven background workers responding to database document lifecycle events (`onCreate`, `onUpdate`) to handle push notifications, metrics aggregation, and cryptographic blockchain anchoring.

3. **Scheduled Cron Workers:** Cloud Pub/Sub workers executing periodic rollups for operational telemetry.

```
Mobile / Web Clients ──> HTTPS Callable Functions ──> Firestore Document Mutation
                                                              │
                                                              ▼ (Event-Driven Trigger)
                                                    Cloud Function Background Worker
                                                              │
                                                  ┌───────────┴──────────┐
                                                  ▼                      ▼
                                          FCM Multicast Push      Web3 0-ETH Anchor
```

2. API Service Catalog

API 1: `createSOS` (Emergency Distress Ingestion)

2.1 Business Meaning & Operational Intent

- **Business Purpose:** Allows a distressed pilgrim or bystander to instantly signal a critical emergency to

- **Business Purpose:** Allows a distressed pilgrim or bystander to instantly signal a critical emergency to state police and medical responders.
- **Business Function:** Establishes the official opening of an emergency incident ticket, capturing the exact spatial coordinates of the victim, filtering out accidental duplicate submissions, and preventing Denial-of-Service spam.
- **Primary Actor:** Devotee / Pilgrim (`pilgrim`)
- **Stakeholder Impact:** Direct determinant of golden-hour medical intervention; triggers simultaneous alerts across police dispatchers and nearby ambulance drivers.

2.2 Technical Implementation Specification

- **Callable Identifier:** `createSOS` (`functions/src/index.ts:20-116`)
- **Invocation Method:** HTTPS Callable Function (`functions.https.onCall`)
- **Region:** `asia-south1` (Mumbai)
- **Authentication:** Mandatory (`context.auth != null`)
- **Input Schema:**

```
json { "latitude": 18.52043, "longitude": 73.85674, "severity": "critical", "idempotencyKey": "e4b3c9a0-4f81-42e1-9311-285b7964b4c2", "description": "Severe chest pain and difficulty breathing near rest camp", "extraDetails": { "bloodGroup": "O+", "batteryLevel": 42 }, "createdAtClient": 1723368400000 }
```

- **Output Schema (Success):**

```
json { "alertId": "sos_7894aBcDeFg123", "status": "pending", "message": "SOS created successfully" }
```

- **Output Schema (Idempotent Duplicate Match):**

```
json { "alertId": "sos_7894aBcDeFg123", "status": "pending", "message": "SOS already exists (duplicate prevented)" }
```

2.3 Business Rules Enforced in API

1. **BR-API-01 (Idempotency):** The engine queries `sosAlerts` for matching `idempotencyKey`. If present, returns the existing record without executing a database write.
2. **BR-API-02 (Rate Limiting):** A device/user is constrained to a maximum of 5 alerts per 60-second sliding window. Exceeding this returns HTTP 429 equivalent (`resource-exhausted`).
3. **BR-API-03 (Geofence Validation):** Coordinates must fall within sovereign Indian bounds ($8.0 \le \text{lat} \le 35.0, 68.0 \le \text{lon} \le 97.5$).

2.4 Error Codes & Exception Contracts

Error Code	HTTP Equivalent	Trigger Condition	Client User Impact
`unauthenticated`	401 Unauthorized	Missing or expired Firebase Auth token.	Forces client to refresh token or re-authenticate.
`invalid-argument`	400 Bad Request	Missing `latitude`, `longitude`, or `idempotencyKey`.	Client displays parameter error dialog.
`resource-exhausted`	429 Too Many Requests	User exceeded 5 SOS calls within 60 seconds.	Displays rate-limit cooldown timer in Marathi.
`failed-precondition`	412 Precondition Failed	Location falls outside Indian geographical boundaries.	Aborts request and flags potential location spoofing.

API 2: `getMedicalCamps` (Geospatial Healthcare Camp Radar)

2.1 Business Meaning & Operational Intent

- **Business Purpose:** Allows pilgrims, field volunteers, and paramedics to locate the closest operational medical camps and inspect available bed capacity.
- **Business Function:** Executes a spatial bounding-box query to convert GPS coordinates into an ordered list of healthcare facilities ranked by physical proximity.
- **Primary Actor:** Pilgrim, Volunteer, EMT Paramedic
- **Stakeholder Impact:** Balances patient intake across rural primary health centers; prevents camp overcrowding during acute heatwave conditions.

2.2 Technical Implementation Specification

- **Callable Identifier:** `getMedicalCamps` (`functions/src/index.ts:122-177`)
- **Invocation Method:** HTTPS Callable Function (`functions.https.onCall`)
- **Region:** `asia-south1`
- **Authentication:** Mandatory (`context.auth != null`)

- **Input Schema:**

  ```
  json { "latitude": 18.3448, "longitude": 74.0298, "radiusKm": 10 }
  ```

- **Output Schema (Success):**

  ```
  json { "camps": [ { "id": "camp_saswad_01", "name": "Saswad Rural Hospital Relief Post", "latitude": 18.3490, "longitude": 74.0320, "geohash": "tek3x8...", "capacity": 30, "currentPatients": 12, "staffCount": 6, "distanceKm": 0.52 }, { "id": "camp_dive_ghat_02", "name": "Dive Ghat Mobile Trauma Unit", "latitude": 18.3800, "longitude": 74.0100, "geohash": "tek3w2...", "capacity": 15, "currentPatients": 14, "staffCount": 4, "distanceKm": 4.31 } ] }
  ```

2.3 Business Rules Enforced in API

1. **BR-API-04 (Active Facilities Only):** Database queries strictly enforce `where('status', '==', 'active')` to prevent routing patients to decommissioned camps.
2. **BR-API-05 (Haversine Distance Pruning):** Any facility returned by the broad Geohash bounding-box query whose computed Great-Circle distance exceeds `radiusKm` is filtered out before sorting.

API 3: `onSOSCreate` (Automated Multicast & Web3 Anchoring Trigger)

3.1 Business Meaning & Operational Intent

- **Business Purpose:** Instantly alerts all on-duty emergency authorities the moment a new distress alert is recorded and anchors an unalterable proof to the blockchain.
- **Business Function:** Asynchronous event handler that queries responder push tokens, dispatches high-priority push notifications, increments live dashboard counters, and creates an immutable cryptographic event snapshot.
- **Primary Actor:** System Event Worker (Internal Cloud Event)
- **Stakeholder Impact:** Eliminates manual radio relay delays; creates an auditable record of the exact second authorities were notified.

3.2 Technical Implementation Specification

- **Trigger Path:** `firestore.document('sosAlerts/{alertId}').onCreate` (`functions/src/index.ts:182-264`)
- **Trigger Event:** Creation of a new document in the `sosAlerts` collection.
- **Downstream Integrations:**
  1. **Firebase Cloud Messaging:** Queries `users` with `userType` $\in$ `['admin', 'police_officer',`

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1. **Firebase Cloud Messaging:** Queries `users` with `userType` $\in$ `['admin', 'police_officer', 'medical_officer']` and calls `admin.messaging().sendEachForMulticast()`.

2. **Singleton Analytics:** Updates `analytics/dashboardMetrics` by executing atomic increment: `activeSOS: FieldValue.increment(1)`.

3. **Web3 Trust Layer:** Computes canonical JSON snapshot, generates SHA-256 hash, and calls `anchorHash(canonicalHash)` to write to Ethereum Sepolia testnet.

4. **Blockchain Proof Collection:** Appends verification record to `blockchainProofs` with `transactionHash`, `canonicalHash`, and `snapshot`.

API 4: `onSOSUpdate` (State Transition Tracking & Proof Versioning)

4.1 Business Meaning & Operational Intent

- **Business Purpose:** Tracks tactical progress as police or medical units respond to, reach, and resolve an emergency, preventing retroactive alteration of incident logs.
- **Business Function:** Detects meaningful status changes (`pending` $\rightarrow$ `acknowledged` $\rightarrow$ `dispatched` $\rightarrow$ `resolved`), increments the event version counter, and anchors an incremental cryptographic proof.
- **Implementation Source:** `functions/src/index.ts:269-312`

API 5: `updateDashboardMetrics` (Scheduled Metric Aggregator)

4.1 Business Meaning & Operational Intent

- **Business Purpose:** Maintains pre-computed, high-speed aggregate statistics for the executive command dashboard without requiring costly full-table scans.
- **Execution Interval:** Scheduled Pub/Sub cron firing every 1 minute (`functions/src/index.ts:318-335`).
- **Technical Operation:** Executes parallel Firestore `.count().get()` aggregations across pending SOS alerts, active medical camps, and registered pilgrims, writing totals to `analytics/dashboardMetrics`.

3. API Security & Secret Management Audit

- **No Hardcoded Secrets:** Review of `functions/src/index.ts` and `functions/src/trustLayer/` confirms that no private API keys, Ethereum private keys, or passwords are hardcoded.
- **Environment Configuration:** Sensitive parameters (Sepolia RPC URL, private key for signing 0-ETH transactions) are injected exclusively via environment variables (`process.env.BLOCKCHAIN_PRIVATE_KEY`, `process.env.SEPOLIA_RPC_URL`).
- **Fail-Safe RPC Handling:** If environment variables or the external RPC endpoint are unavailable, the trust layer gracefully downgrades to `LOCAL_PROOF_ONLY`, ensuring emergency dispatch is never blocked.